

Aryan Jagani

ETL & Pipeline Development · Cloud Data Platforms · Analytics Engineering
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PROFESSIONAL SUMMARY

Data engineer with M.S. in Information Systems (GPA 3.95) with hands-on experience building end-to-end data systems: Python/SQL ETL pipelines, FastAPI and Flask REST services, relational and vector data stores, GCP cloud storage, Dockerized reproducible workflows, and Power BI semantic models. Shipped a geospatial data platform in active use by international research teams and placed 2nd nationally in the NSF HDR machine learning challenge.

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, JavaScript/TypeScript, Bash
Data Engineering: ETL/ELT pipeline design, data ingestion, multi-source harmonization, data quality & QC, schema design, feature engineering, time-series aggregation, Pandas, NumPy
Data Platforms: Databricks, Apache Spark, PySpark, Spark SQL
Data Architecture: Delta Lake, lakehouse architecture, medallion architecture
Data Warehousing: Fact tables, dimension tables, star schema, slowly changing dimensions
Pipeline Development: ETL/ELT, incremental loading, batch processing
Databases & Storage: PostgreSQL, MySQL, SQLite, MongoDB, ChromaDB
Cloud, DevOps & APIs: GCP, Docker, Render, Git/GitHub, CI/CD, Linux, FastAPI, Flask, REST API design, role-based access control, React/Next.js
BI & Visualization: Power BI (DAX semantic modeling, choropleths, KPI dashboards), Tableau, Plotly, Matplotlib, Seaborn, Excel
ML & Practices: scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, LSTM, LangChain/RAG, OpenAI API, model evaluation (ROC-AUC, F1, MCC); reproducible pipelines, BioCompute Objects (BCO), FAIR data principles

PROFESSIONAL EXPERIENCE

Data & Research Engineer (iHARP Fellow) · University of Maryland, Baltimore County Sep 2025 – Present

- Architected and shipped a full-stack geospatial data platform on a FastAPI backend with a SQLite workflow layer, serving browser-based hex-level GeoTIFF classification to international research teams.
- Built ingestion and export pipelines for satellite raster data with annotation assignment tracking, QC review queues, and structured outputs feeding downstream research publication.
- Implemented role-based access control across annotator and reviewer workflows; integrated GCP object storage and ArcGIS basemaps, deployed to Render with environment-based config.
- Engineered Power BI semantic models and dashboards (global choropleths, country profiles, cross-version indicator comparisons) for the UNDP-aligned Nature Relationship Index.

Machine Learning & Data Engineer · GWU HIVE Lab (Mazumder Lab), Remote Jun 2026 – Jul 2026

- Built extraction and harmonization pipelines unifying heterogeneous public multi-omics and PMID-sourced datasets into modeling-ready feature tables with automated quality-control checks.
- Containerized end-to-end analysis workflows in Docker and documented them as BioCompute Objects for FAIR-compliant reproducibility; published scripts and container definitions to GitHub for lab reuse.
- Trained and evaluated scikit-learn risk classifiers on harmonized clinical/genomic features and shipped a Flask demo app for the research symposium.

Graduate Researcher — M.S. Thesis · UMBC (Advisor: Prof. Karen Chen) Sep 2025 – Present

- Designed reproducible Python/Jupyter evaluation pipelines benchmarking GPT-4, Gemini, and Llama with tracked prompt versions, model configurations, and versioned result sets (MLflow-style experiment lifecycle discipline).
- Automated scoring with ROUGE, BLEU, and BERTScore alongside a structured human rubric; ran inter-rater reliability analysis across evaluators.

Artificial Intelligence Intern · Edunet Foundation (AICTE / IBM SkillsBuild), Remote Jun 2023 – Jul 2023

- Built cleaning, preprocessing, and feature-engineering pipelines over health indicator data spanning 150+ countries; benchmarked 12 regression models with hyperparameter tuning and ensembling, reaching 98.5% on the target metric.

PROJECTS

End-to-End ELT Pipeline (Databricks, dbt, Spark SQL, Delta Lake)

- Built an end-to-end ELT pipeline using Databricks, dbt, Spark SQL, and Delta Lake to transform raw data into analytics-ready datasets.
- Implemented bronze, silver, and gold layers following the medallion architecture for ingestion, data cleaning, business transformations, and reporting.
- Developed modular dbt models using source() and ref(), including incremental models, snapshots, macros, and multiple materialization strategies.
- Added automated uniqueness, non-null, accepted-value, and relationship tests to validate data quality and referential integrity.
- Designed fact and dimension tables using star-schema principles and generated dbt documentation and lineage for maintainability.

PDF-to-RAG Ingestion Pipeline (Python, LangChain, ChromaDB, PyMuPDF4LLM, OpenAI) [Link](#) – structure-aware ingestion converting complex scientific PDFs to Markdown while preserving headers, tables, and lists; hierarchical chunking, embedding generation, persistent vector store, and CLI ingestion/retrieval workflows.

Coastal Flood Forecasting Pipeline — NSF HDR Challenge, 2nd Place Nationally (Python, XGBoost, TensorFlow/LSTM) [Link](#) – parsed 70 years of hourly tide-gauge data (1950–2020, 5 stations) from MATLAB datenum format, aggregated to daily grain, and engineered rolling windows for 14-day horizon forecasting; ROC-AUC 0.871, leaderboard F1 0.94.

Live Market Data Dashboard (Flask, Pandas, Render) [Link](#) – ingests live Excel market exports, cleans malformed input (offset headers, duplicate header rows, non-numeric values), and serves a JSON API plus browser dashboard, deployed via render.yaml.

Power BI Analytics Portfolio (Power BI, DAX) [Link](#) – HR attrition model across 1,470 employees surfacing a 16.12% attrition rate by department, role, and age band; Flipmart sales & profit dashboard with segment, ship-mode, and sub-category drilldowns.

EDUCATION

M.S., Information Systems · University of Maryland, Baltimore County | GPA: 3.95 / 4.00 Aug 2026
B.E., Computer Science · Shah and Anchor Kutchhi Engineering College, Mumbai, India May 2024

AWARDS, PUBLICATION & CERTIFICATIONS

- 2nd Place, NSF HDR Scientific-MOOD FAIR Challenge (Year 2) — Coastal Flooding Prediction track, national NSF competition, Mar 2026.
- CogniPoseAI: A Futuristic AI-Enhanced Personal Trainer — IEEE Proceedings, ICSSAS 2023. DOI: 10.1109/ICSSAS57918.2023.10331759
- Certifications — Programming for Everybody (Python), University of Michigan / Coursera; Data Science; LaTeX.